Theorem 1: blowup distances equal optimal transport distances

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Theorem1_BlowupEqualsOT`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper (Theorem 1, p. 3, l. 92; proof in Appendix B, pp. 15–20). For all $1 \le p, q \le \infty$, $\delta^{\bullet}_{p\to q} = \delta^{\mathrm{OT}}_{p\to q}$. Here, for graphs G on V ($|V| = m$) and H on W ($|W| = n$),

$$\delta^{\bullet}_{p\to q}(G,H) = \lim_{\ell\to\infty} \delta_{p\to q}\big(G^{\bullet n\ell}, H^{\bullet m\ell}\big), \qquad \delta^{\mathrm{OT}}_{p\to q}(G,H) = \min_{Q^*\in \mathrm{DS}^*(V,W)} \big\| B_{p\to q}(G,H,Q^*)\big\|_{p\to q} \quad \text{(Eq. (2))},$$

$B_{p\to q}(G,H,Q^*)\big((v,w),(v',w')\big) = Q^*(v,w)^{1/q}\,\big(A_G(v,v') - A_H(w,w')\big)\,Q^*(v',w')^{1-1/p}$ with $0^0 := 0$, and $\delta_{p\to q} = N^{-\kappa} d_{p\to q}$ for graphs of order N, $\kappa = 1/q + 1 - 1/p$ (p. 3, l. 88–90; Appendix B, p. 15, l. 513–514). The paper marks blowups with a bomb symbol, written $\bullet$ here ($G^{\bullet k}$, $\delta^{\bullet}$); the Lean names use `blowup`.

Lean. For all $p, q \in [1,\infty]$ (type `ENNReal`) and all finite simple graphs G on a **nonempty** V and H on a **nonempty** W:

- `GsimOT.theorem1`: `deltaBlowup p q G H = deltaOTPQ p q G H`;
- `GsimOT.theorem1_tendsto`: $\delta_{p\to q}(G^{\bullet n\ell}, H^{\bullet m\ell}) \to \delta^{\mathrm{OT}}_{p\to q}(G,H)$ as $\ell\to\infty$, so the limit defining $\delta^{\bullet}$ *exists* (the paper cites [Gro25] for this, p. 2, l. 44–45);
- `GsimOT.theorem1_of_tendsto`: the paper's argument literally: *if* the sequence has a limit, then $\delta^{\bullet}_{p\to q}(G,H) = \delta^{\mathrm{OT}}_{p\to q}(G,H)$.

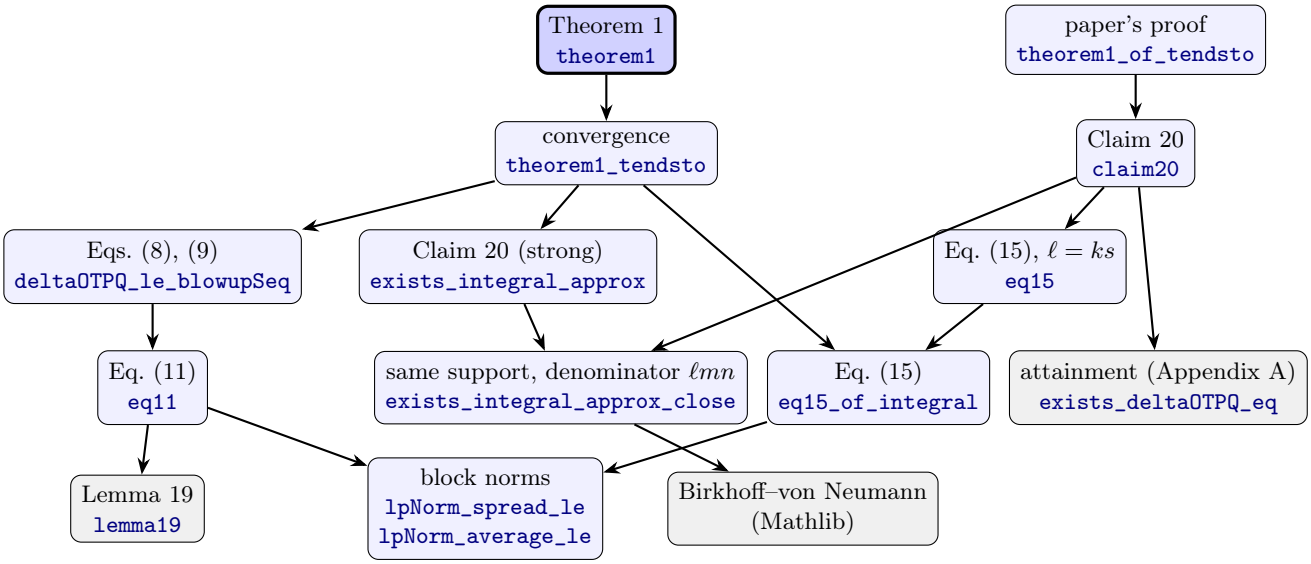
`deltaBlowup` is `limUnder atTop` of `blowupSeq p q G H ell` $= \delta_{p\to q}(G^{\bullet n\ell}, H^{\bullet m\ell})$; `deltaOTPQ` is the infimum over couplings (the minimum is attained, `exists_deltaOTPQ_eq`). The Lean result is therefore the paper's statement plus the existence of the limit, without any cited result.

2 Where the proof lives

File	Content
`Definitions.lean`	blowup $G^{\bullet k}$ on $V\times[k]$ (`blowup`), the sequence `blowupSeq` and `deltaBlowup`.
`BlockNorms.lean`	norm bounds for spread and averaged test vectors over the blocks $J(v,w)$ (proofs of Eqs. (11), (15); Hölder in the form of Eq. (5)).
`LowerBound.lean`	Eqs. (8)–(11): the coupling $Q^* = \lvert J\rvert/N$ induced by a bijection of the blowups, Eq. (10), `eq11` (= Eq. (11)), `deltaOTPQ_le_blowupSeq`.
`UpperBound.lean`	Eq. (15) for every ℓ with $\ell mn\,Q^*$ integral: the bijection π built from the partitions $I(v,w)$, $J(v,w)$, averaged test vectors (`eq15_of_integral`).
`Approximation.lean`	approximation of a coupling by couplings with denominator ℓmn and the same support (`exists_integral_approx_close`), with norm control (`exists_integral_approx`).
`Claim20.lean`	Claim 20 (`claim20`), Eq. (15) for $\ell = ks$ (`eq15`), and the paper's version of the conclusion (`theorem1_of_tendsto`).
`Theorem1.lean`	`theorem1_tendsto`, `theorem1`.
`../Common/OperatorNorm.lean`	ℓ^p norms, $\lVert\cdot\rVert_{p\to q}$, Lemma 19 (`lemma19`).

File	Content
../Common/OTDistances.lean	$\mathrm{DS}^*(V,W)$ (IsCoupling), 0^a (powZ), $B_{p\to q}$ (Bmat), Eq. (2) (deltaOTPQ).
../Common/AlignedDistances.lean	κ, $A^\pi_G - A_H$, $d_{p\to q}$, $\delta_{p\to q}$.
../Common/Attainment.lean	Appendix A: the minimum in Eq. (2) is attained (exists_deltaOTPQ_eq).

3 Proof layout



White boxes: this folder; grey: GsimOT/Common or Mathlib. Not drawn: eq15_of_integral also uses Lemma 19, and theorem1_of_tendsto also uses Eqs. (8), (9). No cited result is assumed.

4 The proof step by step

First direction, Eq. (8) (pp. 15–17, l. 516–567). Fix $\ell \geq 1$, $N = \ell mn$, and a bijection π of the blowups attaining $d_{p\to q}(G^\bullet, H^\bullet)$ (a minimum over a finite set; Lean simply bounds every π). The sets $I(v,w)$, $J(v,w)$ are the blocks of the map $(w,j) \mapsto (v(w,j),w)$ (blockMap, blockCard); $Q^*(v,w) = |J(v,w)|/N$ is couplingOf, a coupling by sum_blockCard_row, sum_blockCard_col, isCoupling_couplingOf. Eq. (10), $M_\pi((w,j),(w',j')) = D(v,w,v',w')$, is alignedDiff_blowup_eq_disc. Eq. (11), $\|M_\pi\|_{p\to q} \geq N^\kappa \|B_{p\to q}(G,H,Q^*)\|$, is eq11: optimal x,y from Lemma 19 (exists_abs_dotProduct_mulVec_eq_opNorm), the spread vectors $\tilde{x},\tilde{y}$ with $\|\tilde{x}\|_p, \|\tilde{y}\|_{q^*} \leq 1$ (lpNorm_spread_le, the paper's computation on p. 16, l. 543–553), and the chain of equalities on p. 17, l. 554–567 (sum_comp_eq_sum_blockCard, div_rpow_self_eq_powZ, powZ_eq_rpow_mul_powZ_div; terms with $|J| = 0$ vanish by $0^0 = 0$). Eqs. (8), (9) follow: deltaOTPQ_le_blowupSeq.

Converse, Eqs. (12), (13) and Claim 20 (pp. 17–18, l. 568–581). Claim 20: for $\varepsilon > 0$ there is a *rational* Q^* with $\|B(Q^*)\| \leq \min \|B\| + \varepsilon$. The paper's proof: take a minimiser Q_0^* (exists_deltaOTPQ_eq, Appendix A), rational $Q_k^* \to Q_0^*$ with $\mathrm{supp}(Q_k^*) = \mathrm{supp}(Q_0^*)$ (exists_rational_seq_supp_eq), entries of $B_{p\to q}$ converge on the fixed support and the norm is continuous (tendsto_opNorm_Bmat_of_supp_eq, via tendsto_powZ_apply, continuous_opNorm). claim20 combines these exactly as the paper does.

The rational sequence is obtained differently from the paper (which uses that the couplings vanishing outside $S = \mathrm{supp}(Q_0^*)$ form a rational polytope, p. 17, l. 574–575): exists_integral_approx_close lifts Q_0^* to the doubly stochastic $mn \times mn$ matrix $R((v,i),f(w,j)) = Q_0^*(v,w)$ (liftM, liftM_mem_doublyStochastic), decomposes it by Mathlib's Birkhoff–von Neumann theorem, rounds the weights to multiples of $1/\ell$ keeping their signs (exists_round_weights), and averages back (avgM, isCoupling_avgM, avgM_liftM). The result has denominator ℓmn, the same support, and is δ-close, for *every* large ℓ.

Eqs. (14), (15) (pp. 18–20, l. 582–638). With k clearing the denominators (`exists_denom`) and $\ell = ks$, $q(v,w) = NQ^*(v,w) \in \mathbb{N}$. The partitions $I(v,w)$ of $[\ell n]$ and $J(v,w)$ of $[\ell m]$ and the glued bijection π are `exists_fin_fun_card_fiber`, `card_fiber_prod`, `exists_blowup_bij`; Eq. (10) holds again. Averaged vectors x, y satisfy $\|x\|_p, \|y\|_{q^*} \le 1$ by Hölder, Eq. (5) (`lpNorm_average_le`); the chain of equalities on pp. 19–20 is `block_sum_eq` inside `eq15_of_integral`, which proves Eq. (15) for every ℓ with $\ell mn\,Q^*$ integral; `eq15` is the case $\ell = ks$.

Conclusion. The paper: $\delta^\bullet = \lim_\ell = \lim_s \delta(G^{\bullet nks}, H^{\bullet mks}) \le \|B\| \le \delta^{\mathrm{OT}} + \varepsilon$ (p. 18, l. 589–590; `theorem1_of_tendsto`, which assumes that the limit exists). `theorem1_tendsto` instead combines Eq. (8) with `exists_integral_approx` and `eq15_of_integral` for *all* large ℓ: every term of the sequence lies in $[\delta^{\mathrm{OT}}, \delta^{\mathrm{OT}} + \varepsilon]$ eventually, so the sequence converges; `theorem1` reads off the limit.

5 Assumptions and trusted base

No hypotheses beyond $1 \le p, q$ and $V, W \ne \emptyset$; no cited results (the existence of the limit, for which the paper cites [Gro25], and the rational approximation of Claim 20 are proved). Only the axioms `propext`, `Classical.choice`, `Quot.sound` are used.

6 Notes on the formalisation

- **Existence of the limit** is proved (`theorem1_tendsto`); the paper cites [Gro25] (p. 2, l. 44–45). The literal argument with this citation as a hypothesis is `theorem1_of_tendsto`.
- **Claim 20**: the paper's argument is followed, but the rational sequence with fixed support is constructed by Birkhoff rounding instead of the rational-polytope argument.
- **Minimiser.** Claim 20 starts from a minimiser; its existence (Appendix A, pp. 14–15, l. 503–510) is formalised (`lowerSemicontinuous_opNorm_Bmat`, `isCompact_setOf_isCoupling`). `theorem1` itself only needs near-minimisers.
- Operator norms are suprema over the unit ball; equal to the paper's maximum over the unit sphere.
- Names: `eq11` is Eq. (11), `eq15_of_integral` is Eq. (15); `lemma19` is Lemma 19.

7 How to check this result

```
lake build GsimOT.Theorem1_BlowupEqualsOT.Theorem1
lake env lean scripts/CheckAxioms.lean     # prints the axioms of GsimOT.theorem1
```

In a Lean file importing the module: `#print axioms GsimOT.theorem1`, `#print axioms GsimOT.theorem1_of_tendsto`. The full table of correspondences is `Theorem1_correspondence.pdf` in this folder.